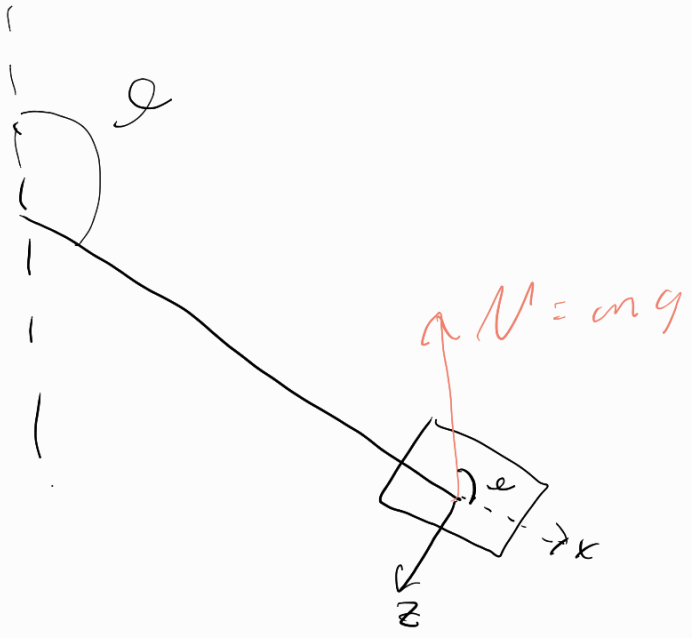
# Problem 1

$$\frac{d}{dt}\begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \\ \lambda \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ k_3 & 0 & 0 & 0 & 0 \end{bmatrix}}_{A} \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \\ \lambda \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & k_1 \\ 0 & 0 \\ k_t/c_2 & 0 \\ 0 & 0 \end{bmatrix}}_{B} \begin{bmatrix} \tilde{V}_s \\ V_\lambda \end{bmatrix}$$




$$y = Cx$$

$$f_n = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \qquad f_b = R_\theta(\theta) R_\lambda(\lambda) f_n$$

$$R_\theta(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

$$R_\psi(\psi) = \begin{bmatrix} \cos(\psi) & \sin(\psi) & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & \sin(\phi) \\ 0 & -\sin(\gamma) & \cos(\phi) \end{bmatrix}$$

$$f_b = R_\theta R_\psi f_n$$

$$f_b = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & \gamma \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\psi) & \sin(\psi) & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ 0 & \gamma & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \gamma & \cos(\phi) & \sin(\phi) \\ 0 & -\sin(\phi)\cos(\phi) & -\ln q \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \end{bmatrix}$$

$$f_b = \begin{bmatrix} \cdot & \cdot & - \\ & & \end{bmatrix} \cdot \begin{bmatrix} \cos(\psi) & \sin(\phi) & 0 \\ -\sin(\psi) & \cos(\phi) & \gamma \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -\sin(\phi) \\ -\cos(\phi \end{bmatrix}, \ln q$$

$$f_b = \begin{bmatrix} \cos(\varphi) & 0 & \sin(\varphi) \\ 0 & 1 & 0 \\ -\sin(\varphi) & 0 & \cos(\varphi) \end{bmatrix} \cdot \begin{bmatrix} -\sin(\alpha)\sin(\beta) \\ -\cos(\alpha)\sin(\beta) \\ -\cos(\beta) \end{bmatrix} \cdot mg$$

$$f_b = \begin{bmatrix} -\cos(\varphi)\sin(\alpha)\cos(\beta) - \cos(\varphi)\cos(\beta) \\ -\cos(\alpha)\sin(\beta) \\ \sin(\varphi)\sin(\alpha)\sin(\beta) - \cos(\varphi)\cos(\beta) \end{bmatrix} \cdot mg$$

$$f_b = \begin{bmatrix} \sin(\theta) \\ -\sin(\phi)\cos(\theta) \\ \cos(\phi)\cos(\theta) \end{bmatrix} mg$$

$$a_x = \sin(\theta)\, g$$

$$a_y = -\sin(\phi)\cos(\theta)\, g$$

$$a_z = \cos(\phi)\cos(\theta)\, g$$

$$\frac{a_y}{a_z} = -\tan(\phi)$$

$$a_y^2 + a_z^2 = \cos(\theta)^2 g^2$$

$$\frac{a_y^2 + a_z^2}{a_x^2} = \frac{\cos(\theta)^2}{\sin(\theta)^2}$$

$$t_{ns}(P) = \frac{dx}{\sqrt{a_y^2 + a_z^2}}$$

# Problem 4

C:

- Kyn og kvinna
- Akkurat núll
- Akkurat núll með karl aðal valkvæði
- Rauður